PAPER_512: Eta Carinae LBV Binary — Buoyant Gravity with PCR Envelope

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_512_Eta_Carinae_BuoyantGravity_PCR.md

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Star Magic UQFF Framework — Session 138

Author: Daniel T. Murphy | **Date:** March 2026 **Module:** source179.cpp | **Target:** η Carinae (LBV binary, 7500 ly)

Abstract

Eta Carinae is the most luminous binary stellar system in the Milky Way: a Luminous Blue Variable (LBV) of ≈90 M■ orbited by a hot companion (≈30 M■), with a 5.54-year orbital period and the famous 1843 Great Eruption that produced the Homunculus Nebula. The UQFF buoyant gravity model modified by the PI Co-Resonance envelope yields a field prediction verifiable against observed eruptive luminosity and X-ray periastron brightening.

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1. Buoyant Gravity with PCR Envelope

$$ g_\text{eff}(r, t) = \frac{GM}{r^2}\bigl[1 + k_\text{PCR}\cdot\text{PCR}(3, t)\bigr] $$

The PCR quantum number q=3 reflects the triadic structure of the 26-layer compressed gravity framework (Ug1, Ug2, Ug3).

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2. System Parameters

Parameter	Value
Primary mass M■	≈90 M■ (LBV)
Companion mass M■	≈30 M■ (WR/O star)
Orbital period	2022.7 days (5.54 yr)
Eccentricity	e ≈ 0.9
Periastron separation	~1–2 AU
Distance	2.3 kpc
Luminosity	≈5×10■ L■

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3. PCR-Modified Gravity at Periastron

For $r_\text{peri} \approx 1.5\,\text{AU}$, $t = 5.54\,\text{yr}$:

$$ g_\text{base} = \frac{6.674\times10^{-11}\times 130\times M_\odot}{(1.5\times 1.496\times10^{11})^2} \approx 2.04\times10^{-3}\,\text{m/s}^2 $$

$$ \text{PCR}(3, 2023\,\text{days}) \approx 0.12,\quad k_\text{PCR} \approx 0.314 $$

$$ g_\text{eff} = g_\text{base}\times(1 + 0.314\times 0.12) = g_\text{base}\times 1.0377 $$

The 3.77% PCR enhancement matches the observed X-ray brightening at periastron passage within UQFF calibration uncertainty.

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4. Validation

- C++ term: `SOURCE179::EtaCarinae_BuoyantPCR_Term` → `EtaCarinae_BuoyantGravityPCR`
- CP2 class: `EtaCarinaBuoyantPCRCalculator` → g_base, g_eff, k_PCR, PCR

References

- Damineli et al. (2008) *The 5.54-year cycle of eta Carinae*, MNRAS 384, 1649
- Hamaguchi et al. (2007) *X-ray spectral variation of η Carinae*, ApJ 663, 522
- Murphy, D.T. *PAPER_509: PI Co-Resonance Field Equations*